

Bengali ASR: six models on FLEURS, one GPU

920 FLEURS bn_in TEST utterances · RTX A5000 · greedy, batch 1, no external LM · all six measured in one sitting

Model	WER	95% CI (clustered)	CER	ms/clip	Params
Conformer Large (hishab)	14.37%	[13.11-15.73]	4.48%	77	121.5M
Whisper Medium	16.23%	[15.05-17.45]	5.09%	1,866	763.9M
Qwen3-ASR + Bengali adapter	16.54%	[15.43-17.68]	4.68%	9,728	2.04B (38M adapter)
► ehzawad FastConformer-CTC	17.12%	[15.95-18.34]	5.12%	83	115.6M
Wav2Vec2	20.64%	[19.46-21.89]	6.02%	64	315.5M
hishab FastConformer	36.22%	[33.88-38.72]	9.05%	79	115.6M

Test split only. The two ehzawad models were trained on corpora that include the FLEURS train split, and 88 of the 150 validation-split sentences also occur there; the 402 validation recordings are therefore excluded for every model and the table is the 920-utterance test split.

Reading this table

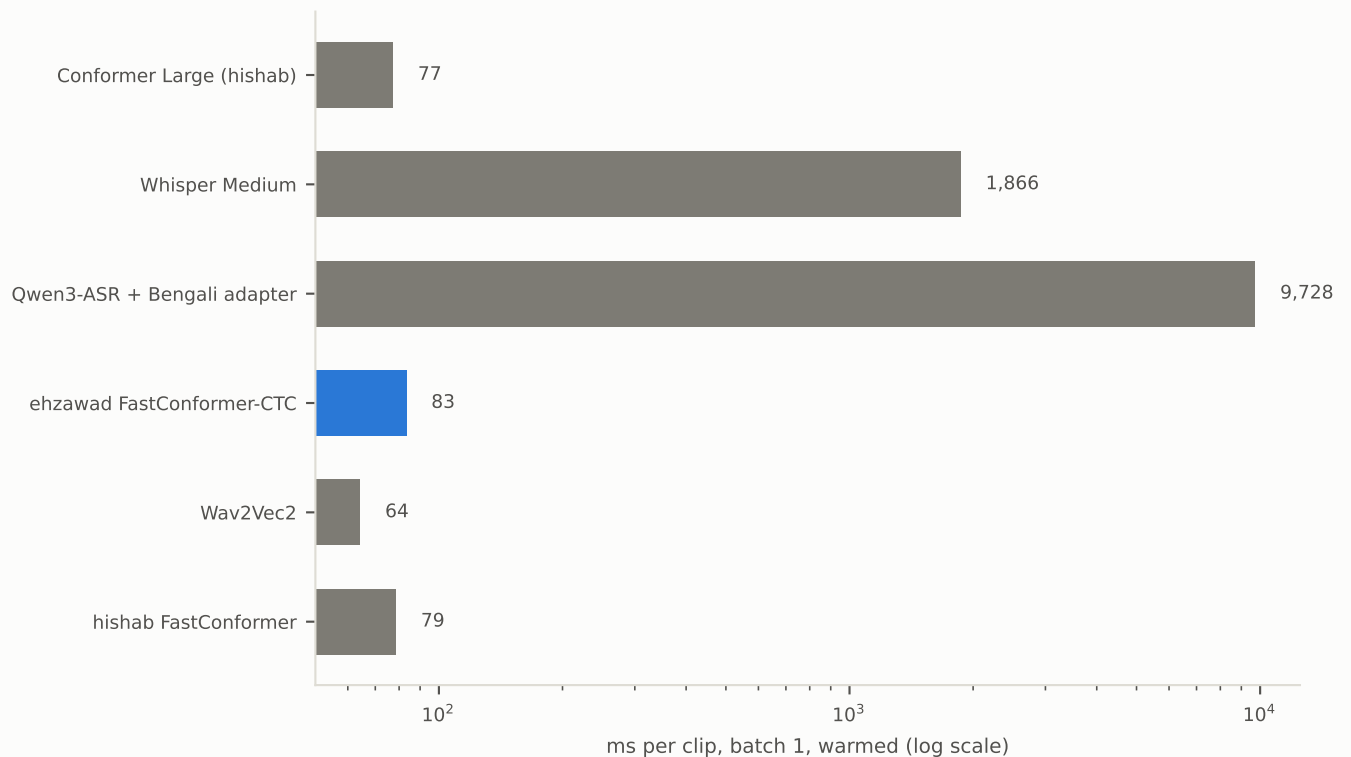
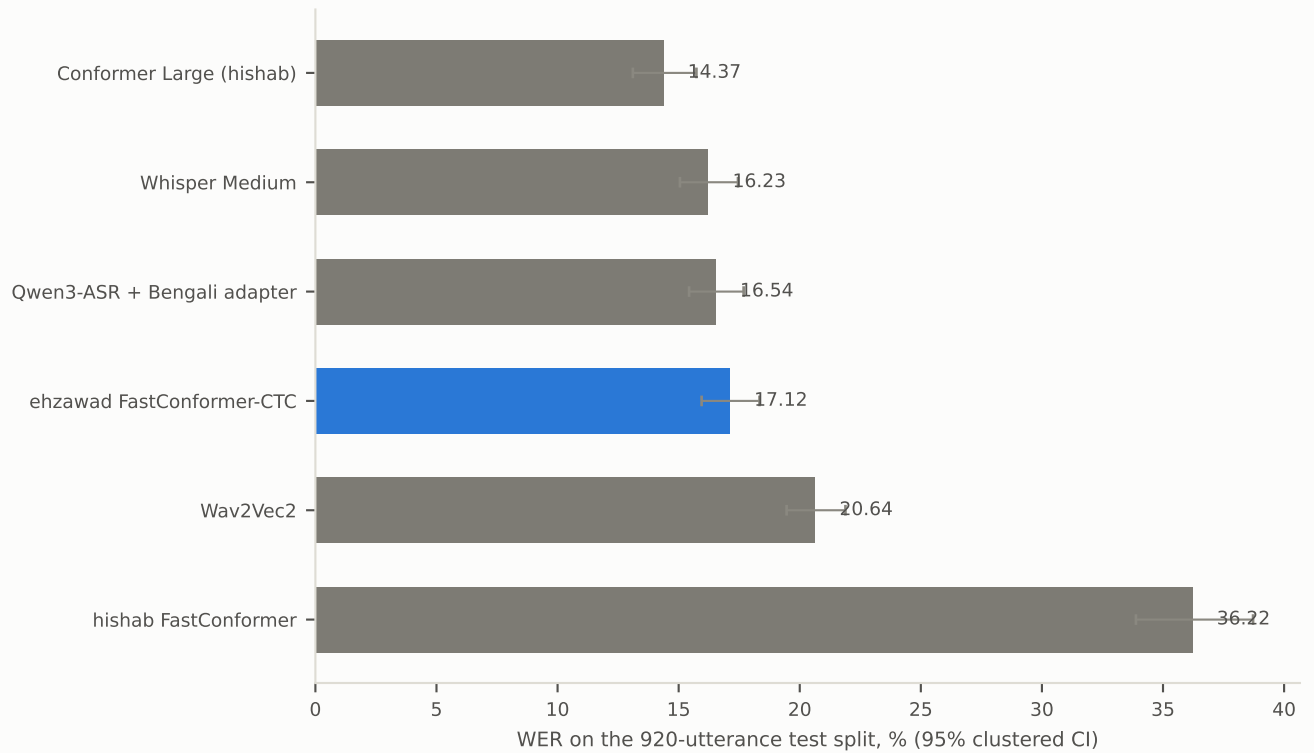
ehzawad/stt_bn_fastconformer_ctc is a FastConformer-CTC large (115.6M parameters, 8x subsampling, 1,024-piece Bengali BPE) trained on 1,692 hours of human-supervised Bengali, CC-BY-SA-4.0. The other five models are public checkpoints from hishab, SayedShaun (mirrors of tugstugi's Whisper and qdv206's wav2vec2 competition models) and ehzawad (a Bengali LoRA adapter on Qwen3-ASR-1.7B: 2.04B parameters, of which 38M were trained).

Paired sentence-clustered deltas for the highlighted model (10,000 draws):

- vs Qwen3 adapter +0.58 pp [-0.24, +1.40]
- vs Whisper Medium +0.89 pp [+0.27, +1.50]
- vs Conformer Large +2.74 pp [+2.01, +3.46]

Not distinguishable from the Qwen3 adapter at 120x its speed; under a point behind Whisper Medium at 22x its speed; 2.7 points behind Conformer Large.

FLEURS test: accuracy and speed side by side



The two orderings are not the same ordering. The three NeMo CTC models sit within 10% of each other in speed; Whisper Medium costs ~22x and the Qwen3 adapter ~120x per clip at batch 1 for their accuracy.

Beyond FLEURS: official test splits of four other corpora

Frozen seeded 1,000-utterance subsets of the OFFICIAL test splits · one model at a time · same backends and scorer

Set	official rows	sampled	audio	ref. words	stratified by
Vaani test	12,695	1,000	1.26 h	10,082	29 state/district strata
SPRING-INX R1	1,879	1,000	2.54 h	19,706	Latin / no-Latin reference
SPRING-INX R2	1,872	1,000	2.61 h	20,353	Latin / no-Latin reference
Kathbath known	2,806	1,000	1.79 h	10,801	speaker gender
Kathbath unknown	1,783	1,000	1.81 h	10,887	speaker gender

Model (WER %)	Vaani	SPRING R1	SPRING R2	Kathbath kn	Kathbath unk	flags
► ehzawad FastConformer-CTC	24.66	39.24	38.41	13.38	13.43	N N N S S
Whisper Medium	26.25	39.96	39.70	9.69	9.79	U U U U U
Conformer Large (hishab)	33.10	42.46	42.24	9.33	8.79	N N N S S
hishab FastConformer	28.71	46.47	45.08	25.81	29.87	U U U U U
Wav2Vec2	31.26	47.28	47.33	19.08	18.80	U U U U U

flags, per set in the column order: N = corpus not in the model's declared training data; S = trained on a sibling split of the same corpus (Kathbath train; Conformer Large also its validation split); U = training provenance not documented. None of these means "verified clean". The highlighted model's Vaani and SPRING hypotheses come from its earlier evaluation of the full official splits (same greedy decoding, batched), scored here on exactly these utterances; they are not re-timed. Its Kathbath rows are batch-1 decodes here. The Qwen3 adapter is not in this comparison (~10 s per clip at batch 1).

What the table says

On the two corpora nobody trained on, the FLEURS ranking does not hold. Conformer Large - the FLEURS leader at 14.4% - is last on Vaani (33.1%) and fourth on SPRING; its training list is Bangladesh-centric and these are Indian-district (Vaani) and IIT-Madras conversational (SPRING) recordings. ehzawad/stt_bn_fastconformer_ctc leads Vaani and both SPRING sets; Whisper Medium is second on all three at ~15x the latency.

On Kathbath the order flips back: Conformer Large (9.3 / 8.8%) and Whisper (9.7 / 9.8%) lead and the highlighted model is third (13.4 / 13.4%). Conformer Large trained on Kathbath validation as well as train, Whisper's provenance for Kathbath is unknown (a U cell can hide training overlap), and the highlighted model saw only the train split. Known- vs unknown-speaker Kathbath differ by under a point for every model except hishab FastConformer. Sentence-clustered 95% intervals for every cell are in the README and summary.

Code-switching, latency, and how the runs were controlled

SPRING-INX: WER on utterances whose reference has no Latin-script character / has Latin-script characters (R1: 499 / 501; R2: 542 / 458), and hypotheses containing any Latin script (of 1,000)

Model	R1 no-Latin / Latin	R2 no-Latin / Latin	hyps with Latin R1 / R2
► ehzawad FastConformer-CTC	35.39 / 41.85	35.14 / 41.12	0 / 0
Whisper Medium	35.92 / 42.71	36.67 / 42.22	0 / 0
Conformer Large (hishab)	40.03 / 44.11	39.53 / 44.49	0 / 0
hishab FastConformer	47.32 / 45.90	43.66 / 46.26	0 / 0
Wav2Vec2	43.99 / 49.51	44.39 / 49.76	0 / 0

The NeMo models' Bengali-only vocabularies cannot emit a Latin word; Whisper and wav2vec2 could in principle, but none of the five produced a single Latin-script hypothesis in 2,000 utterances, so every Latin reference word is an error for every model and the Latin half separates the models less than the Latin-free half does. There the highlighted model (35.4 / 35.1%) and Whisper (35.9 / 36.7%) are within each other's intervals.

ms per clip, batch 1, warmed, default runtime (N/A = stored hypotheses, not timed here)

Model	Vaani	SPRING R1	SPRING R2	Kathbath kn	Kathbath unk
► ehzawad FastConformer-CTC	N/A	N/A	N/A	72	70
Whisper Medium	919	1,454	1,494	1,014	1,041
Conformer Large (hishab)	74	75	76	70	71
hishab FastConformer	72	74	73	70	71
Wav2Vec2	33	49	51	35	35

Protocol. Every set is a frozen subset (seed 20260915; proportional stratification; per-file SHA-256 verified before decoding). One model resident at a time on the A5000; no other process used the card during any run. Manifests: multiset/; raw predictions and per-run GPU state: outputs_multiset/; harness bench_run_set.py (bench_run.py's backends unchanged); scorer score_multiset.py. Reproduce the scoring from the committed predictions.

Evaluation data: FLEURS (Google, CC-BY-4.0); Vaani transcription part (ARTPARK-IISc, CC-BY-4.0); SPRING-INX R1/R2 (SPRING Lab, IIT Madras, CC-BY-4.0); Kathbath (AI4Bharat IndicSUPERB, CC0). Models: hishab (Conformer Large, FastConformer); SayedShaun (Whisper Medium mirror of tugstugi's competition model; wav2vec2 mirror of qdv206's); ehzawad (FastConformer-CTC, Qwen3 adapter). Author of this benchmark: Emrul Zawad.